

LECTURE 13

COMP(1511|1911) 25T2



SOFIA DE BELLIS

MULTI-FILE PROJECTS
LINKED LISTS DEMO PROGRAM

TODAY

Multi-file Projects

Multi-File Demo Program

Linked List Rehash Kahoot

Linked Lists Demo Program



Assignment 2 released



<https://discourse02.cse.unsw.edu.au/25T2/COMP1511/t/assignment-2-released/507>

Wk 8 Revision Sessions



<https://discourse02.cse.unsw.edu.au/25T2/COMP1511/t/week-8-revision-sessions/521>



MULTI-FILE PROJECTS



WHAT ARE THEY?

- Big programs are often spread out over multiple files. There are a number of benefits to this:

1

Improves readability
(reduces length of program).

2

You can separate code by
subject (modularity).

3

Modules can be written
and tested separately.

- So far we have already been using the multi-file capability. Every time we `#include`, we are actually borrowing code from other files.
- We have been only including C standard libraries.



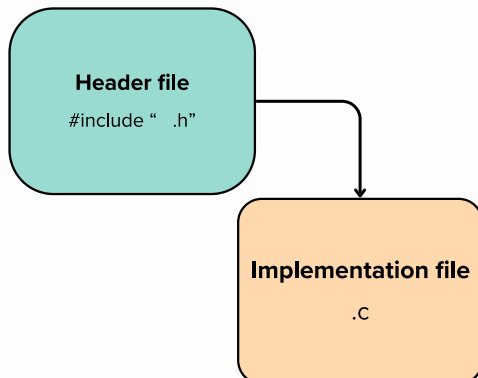
MULTI-FILE PROJECTS

WHAT ARE THEY?

- You can also `#include` your own libraries!
- This allows us to join projects together.
- It also allows multiple people to work together on projects out in the real world.
- We will also often produce code that we can then use again in other projects
 - This is all that the C standard libraries that we have been using are - functions that are useful in multiple instances.

MULTI-FILE PROJECTS

.H AND .C



- In a multi file project we might have:
 - (multiple) header file - this is the .h file that you have been using from standard libraries already
 - (multiple) implementation file - this is a .c file, it implements what is in the header file.
- Each header file that you write, will have its own implementation file
- a main.c file
 - this is the entry to our program
 - contains the main function
 - we try and have as little code here as possible

MULTI-FILE PROJECTS

THE HEADER FILE

Header file

#include " .h"

- Typically contains:
 - function prototypes for all functions that will be implemented in the implementation file
 - comments that describe how the functions will be used
 - #defines
 - enum definitions
 - the file basically SHOWS the programmer all they need to know to use the code
 - NO RUNNING CODE
 - This is like a definition file

MULTI-FILE PROJECTS

THE IMPLEMENTATION FILE

Implementation file

.c

This is where you implement the functions that you have defined in your header file.

MAIN.C

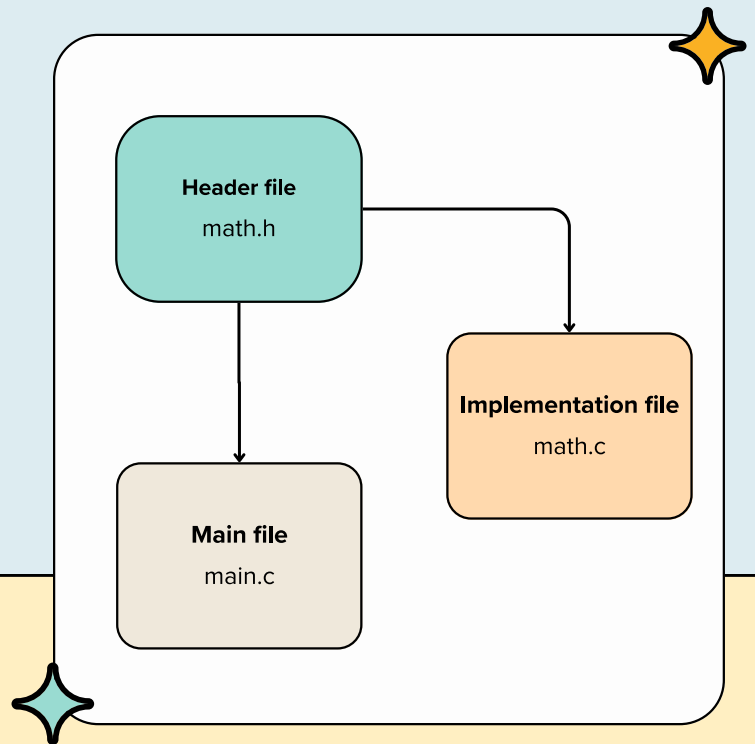
The entry point for your program, where you call functions from that may exist in other modules.

MULTI-FILE PROJECTS

CODE DEMO

We will have three files:

- Header file - math.h
 - #include "math.h"
- Implementation file - math.c
- Main file - main.c
 - #include "math.h"



MULTI-FILE PROJECTS

COMPILING

To compile a multi file project, you list any .c files you have in your project. In the case of our example, we have maths.c and main.c files.

```
sofiad@vx13 ~$ gcc maths.c main.c -o maths
sofiad@vx13 ~$ ./maths
The area of the rectangle is 13.950000
The sum of the two numbers is 12
sofiad@vx13 ~$
```

The program will always enter the main.c, so there should only be one main.c when compiling.

KAHOOT

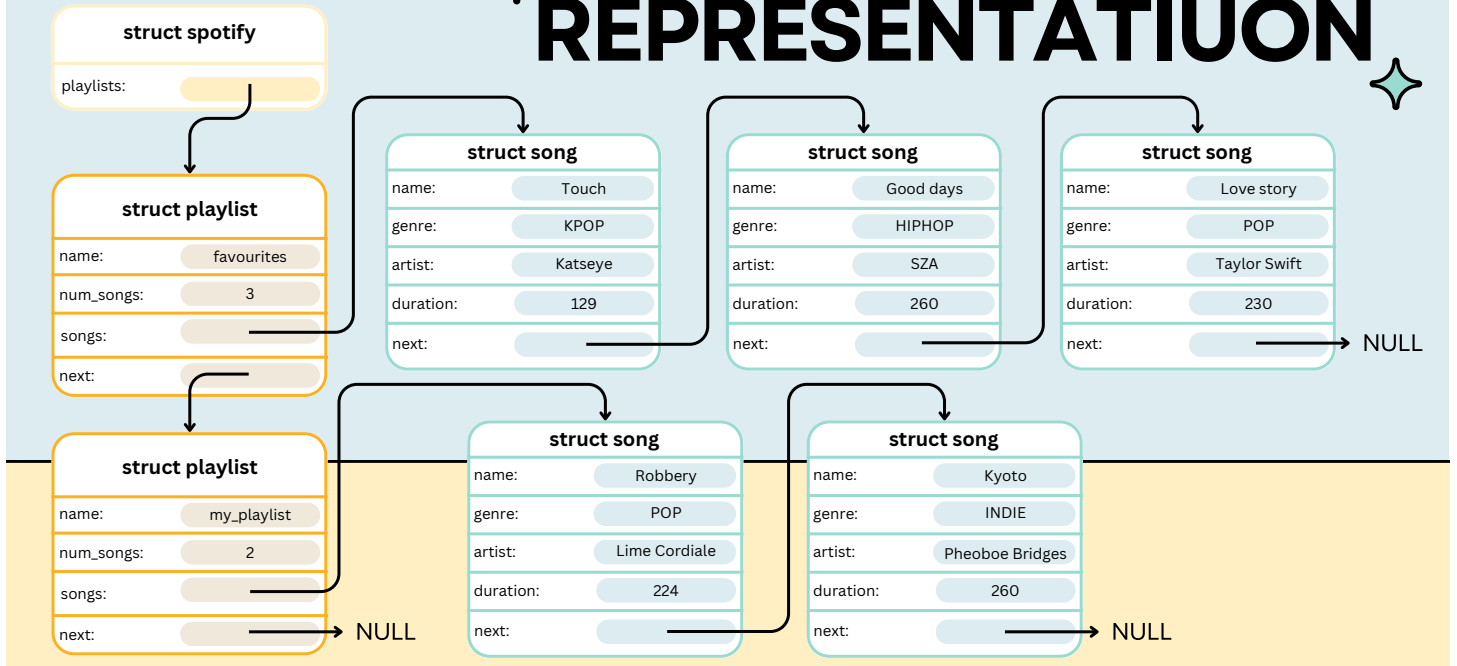
LINKED LISTS DEMO PROGRAM

PLAYLIST MANAGER

You're managing a simple Spotify simulator where each playlist is stored as a node in a linked list, Spotify and each song is stored as a node in a linked list for each playlist.

- Files:
 - spotify.c
 - spotify.h
 - main.c
- Tasks:
 - initialise spotify
 - add/remove playlist
 - add/remove song
 - playlist duration
 - print spotify
 - delete playlist/song/spotify
 - move to top
 - merge
 - ???

VISUAL REPRESENTATION



LIVE CODE TIME

LINKED LISTS CHECKLIST

- ✓ Are you drawing diagrams are you code? - Draw, Code, Repeat!
(It's so much easier to debug this way!)
- ✓ Have you considered all the possible cases we can operate in? Here we some that may apply:
 - How many nodes are in the list?
 - Empty list
 - Only one node
 - More than one/many nodes in the list
 - Which node/where in the list do we want to operate?
 - At the head
 - Between two nodes
 - At the tail

THANK YOU



<https://forms.office.com/r/5x6dail5De>